Lecture 6 - 03/06

Thursday, June 4, 2026 2:56 PM

last time we saw -

$$\begin{cases} \dot{x} = x(a - by) \\ \dot{y} = y(-c + ex) \end{cases}$$

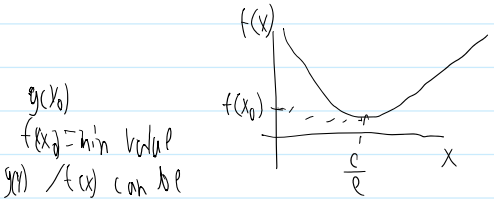
$$\frac{\dot{y}}{\dot{x}} = \frac{y(-c+ex)}{x(a-by)}$$

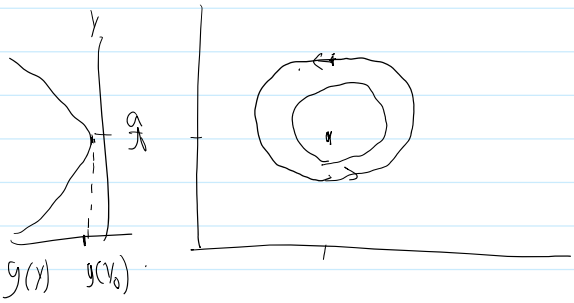
$$\dot{y}\left(\frac{a}{y} - b\right) = \dot{x}\left(-\frac{c}{x} + e\right)$$

integrating both sides gives us

$$K = (ex - c\ln x) + (by - a\ln y) = f(x,y) = \text{const}$$

$f(x)$ $\quad$ $g(y)$





$$K' = K - K_0 = K - f_0 - g_0$$

$$m\ddot{x} = F(x)$$

$$\begin{cases} \dot{x} = v \\ \dot{v} = \frac{1}{m}F(x) \end{cases}$$

$$m\ddot{x} = F(x)$$
$$m\ddot{x}\dot{x} = \dot{x}F(x)$$
$$m\ddot{x}\dot{x} - \dot{x}F(x) = 0$$
$$\frac{d}{dt}\left[\frac{1}{2}m\dot{x}^2 - \int_{x_0}^{x} F(s)\,ds\right] = 0$$

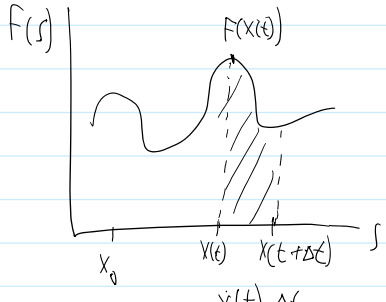
so this is conserved over time, we call it energy

$$E = \frac{1}{2}m\dot{x}^2 - \int_{x_0}^{x} F(s)\,ds = \frac{1}{2}m\dot{x}^2 + V(x)$$

notice $V(x)$ depends on x_0

$$V(x) = -\int_{x_0}^{x} F(s)\,ds$$

$$\lim_{\Delta t \to 0} \frac{\int_{x_0}^{x(t+\Delta t)} F(s)\,ds - \int_{x_0}^{x(t)} F(s)\,ds}{\Delta t} = \lim_{\Delta t \to 0} \int_{x(t)}^{x(t+\Delta t)} F(s)\,ds$$



for Δt small enough, this will be $F(x(t)) \cdot \dot{x} \cdot \Delta t$

the value at $x(t)$ $\quad$ the slope there

for our case, $\dot{x}F(x) \overset{?}{=} \int_{x_0}^{x} F(s)\,ds$

$$\begin{cases} \dot{x} = y & = f(x,y) \\ \dot{y} = x - x^3 & = g(x,y) \end{cases}$$

FP: $y = 0$, $x = 0, \pm 1$

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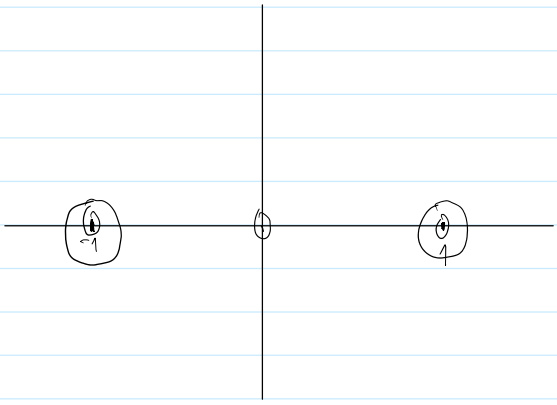
$$DF(x,y) = \begin{pmatrix} 0 & 1 \\ 1-3x^2 & 0 \end{pmatrix}$$

$$DF(0,0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \varphi(\lambda) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 1$$

$\lambda_1 = \pm 1$ saddle

$$DF(\pm 1, 0) = \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix} \qquad \varphi(\lambda) = \begin{vmatrix} -\lambda & 1 \\ -2 & -\lambda \end{vmatrix} = \lambda^2 + 2$$

$\lambda_{1,2} = \sqrt{-2} \pm i\sqrt{2}$ Centers



we could analyze saddle to find stable and unstable manifolds but will do something else, will use energy function

$\dot{x} = v$

$\dot{v} = \frac{1}{m} F(x)$

$E = \frac{m\dot{x}^2}{2} + V(x)$

$V(x) = -\frac{1}{m}\int_{x_0}^{x} F(x)\,dx$

will take $m = 1$?

$m=1, \dot{x}=y$

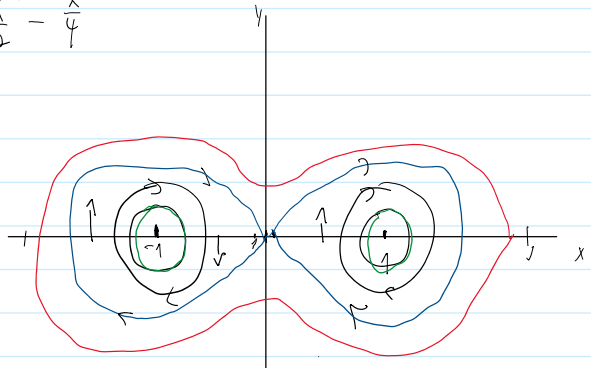
why ?? $\int \dot{y} = \int x - x^3 = \frac{x^2}{2} - \frac{x^4}{4}$

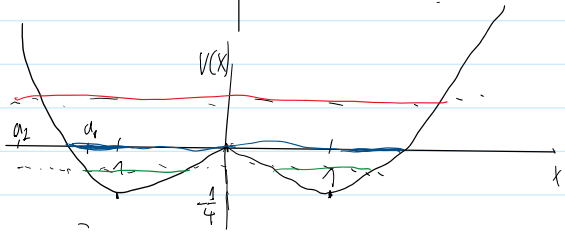
$\dot{E} = 0$

$E = \frac{m\dot{x}^2}{2} - \frac{1}{m}\int_{x_0}^{x} f(x)\,dx = \frac{y}{2} - \int_{x_0}^{x} F(x)\,dx = \frac{y}{2} - \frac{x^2}{2} - \frac{x^4}{4}$

$V(x) = -\frac{x^2}{2} + \frac{x^4}{4}$

blue is homoclinic
it leaves FP and returns to it





if we have $x_0 = a_1$, then $V(a_1)$ will be its max potential energy.
as it goes in the cycle, it slides down in V
at $x=-1$, V will be min, kinetic energy will be max.
it will roll around in the well

very close to 0, it will also cycle and it cannot cross the $y=0$ line on V, since energy is conserved

if we have $x_0 = a_2$, at $x=0$, it still has speed

this is called double well potential

for pendulum

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\sin x \end{cases}$$

$$E(x(t), y(t)) = \frac{y^2}{2} + 1 - \cos x$$

$$V(x) = -\int_0^x -\sin s \, ds = -\cos \Big|_0^x = -\cos x + 1$$

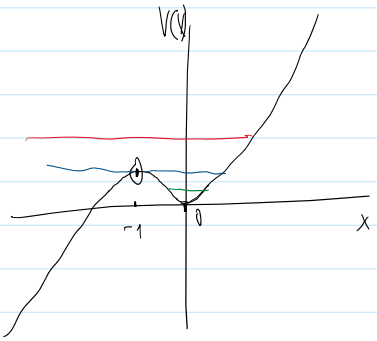
$(0,0)$ should be stable, because it is a min

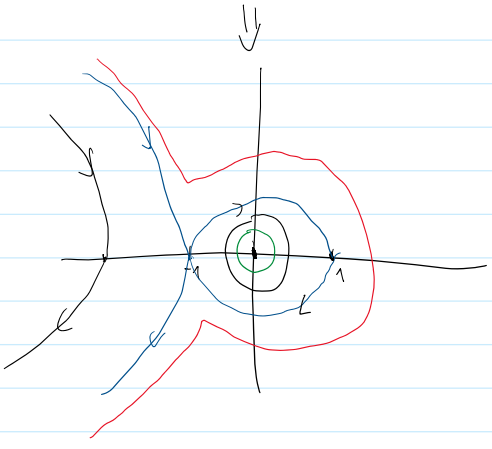
blue is a hetroclinic orbit, it leaves one FP and stops at another

$$\begin{cases} \dot{x} = y \\ \dot{y} = -(x + x^2) \end{cases}$$

$$E = \frac{m v^2}{2} + V(x)$$

$$V(x) = -\int_0^x -s - s^2 \, ds = \frac{x^2}{2} + \frac{x^3}{3}$$





blue is stable manifold
homoclinic orbit

back to the Pendulum - lets add damping

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\sin x - b y \end{cases}$$

$$E = \frac{y^2}{2} + 1 - \cos x$$

$$\dot{E} = y\dot{y} + \sin x \cdot \dot{x} = y(-\sin x - by) + \sin x \cdot y = -by^2 \leq 0$$

so energy is not constant

definition: for dynamical system $\dot{\underline{x}} = f(\underline{x})$, with FP $\underline{x}^*$ if $\exists L(\underline{x})$ s.t.

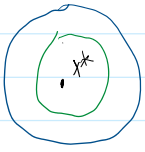
$L(\underline{x}) \geq L(\underline{x}^*)$. for x close to x^*
$\dot{L}(x) \leq 0$

then this is a weak lyapunov function

if $\dot{L}(x) < 0$, its strong

any energy is a WLF

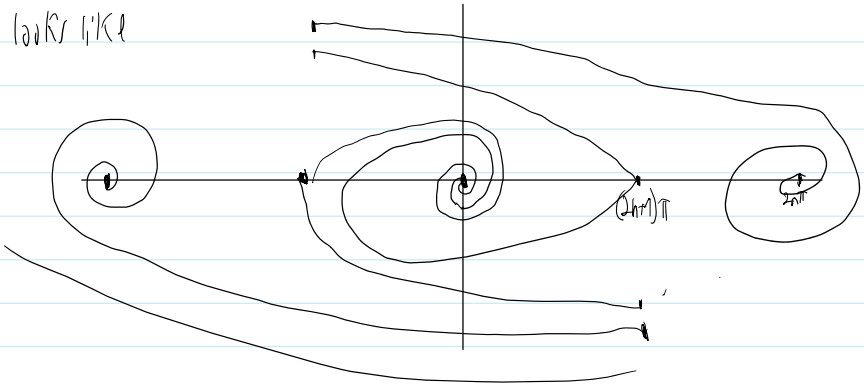
define: FP x^* is lyapunov stable if for a small enough area around x^* all trajectories that start in that area, cannot leave another area
any x_0 in green will never leave blue



1) if you find a WLF in neighborhood of FP, this FP is lyapunov stable
- idea, if $\dot{L}(x) \leq 0$, $L(x)$ cannot increase, so it is stuck in that area

2) if you find a SLF " " " " , this FP is asymptotically stable

So the damped pendulum looks like

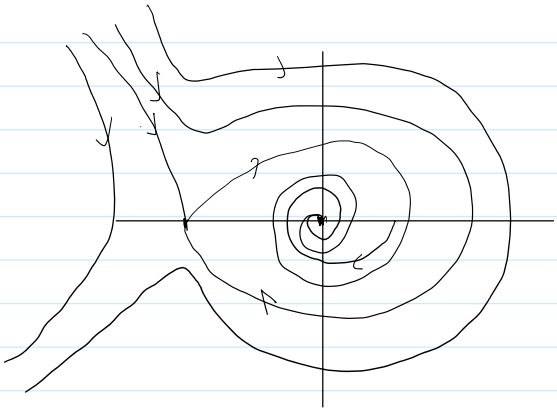


$\Sigma_u : \{ \dot{L}(x) = 0 \mid x \in U \}$

another example, lets damp

another example, lets damp

$$\begin{cases} \dot{x} = v \\ \dot{y} = -(x + x^2) - by \end{cases}$$



$$E = \frac{mv^2}{2} + V(x)$$

$$V(x) = -\int_0^x -s - s^2 \, ds = \frac{x^2}{2} + \frac{x^3}{3}$$

how

$$\dot{E} = -by$$

its a WLF

Lyapunov function is a special case of a test function